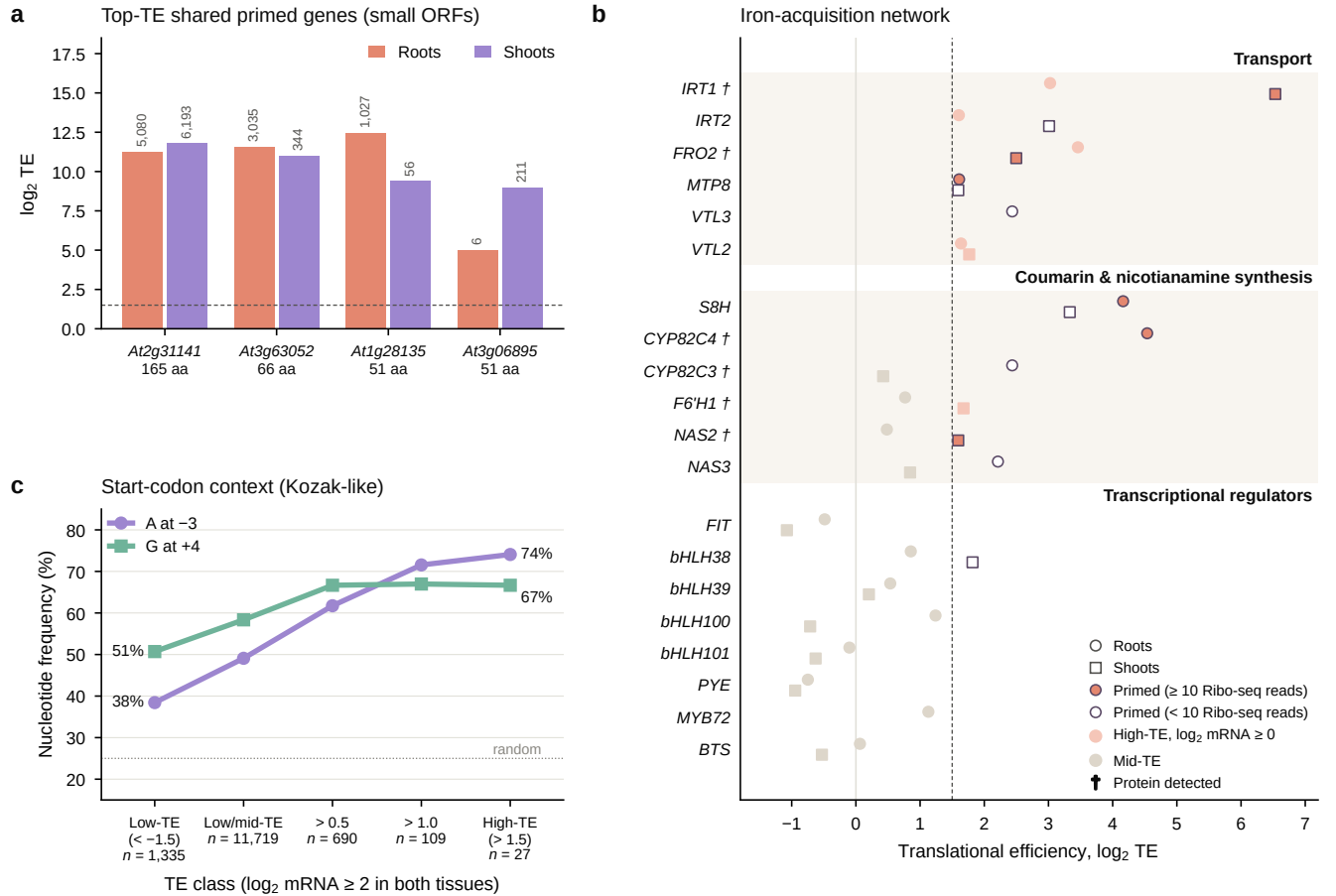
a
Top-TE shared primed genes (small ORFs)
Roots
Shoots
log2 TE
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
5,080
6,193
3,035
344
1,027
56
6
211
At2g31141
165 aa
At3g63052
66 aa
At1g28135
51 aa
At3g06895
51 aa
b
Iron-acquisition network
Transport
IRT1 †
IRT2
FRO2 †
MTP8
VTL3
VTL2
Coumarin & nicotianamine synthesis
S8H
CYP82C4 †
CYP82C3 †
F6'H1 †
NAS2 †
NAS3
Transcriptional regulators
FIT
bHLH38
bHLH39
bHLH100
bHLH101
PYE
MYB72
BTS
Roots
Shoots
Primed (≥ 10 Ribo-seq reads)
Primed (< 10 Ribo-seq reads)
High-TE, log2 mRNA ≥ 0
Mid-TE
Protein detected
−1
0
1
2
3
4
5
6
7
Translational efficiency, log2 TE
c
Start-codon context (Kozak-like)
A at −3
G at +4
Nucleotide frequency (%)
80
70
60
50
40
30
20
74%
67%
51%
38%
random
Low-TE
(< −1.5)
n = 1,335
Low/mid-TE
n = 11,719
> 0.5
n = 690
> 1.0
n = 109
High-TE
(> 1.5)
n = 27
TE class (log2 mRNA ≥ 2 in both tissues)